
4.2 Rhinosinusitis

4.2.1 Acute Rhinosinusitis

◆ **Key Features**

- Rhinosinusitis is an inflammatory condition of the nose and sinuses.
- By definition of acute, signs and symptoms last less than one month.
- The condition resolves with treatment; inadequate treatment may lead to disabling chronic disease.

Acute bacterial rhinosinusitis is frequently managed by primary care providers. The otolaryngologist often sees patients who are inadequately treated or who have recurrent disease. Aggressive care is necessary with immunocompromised patients. Collection of mucopus during nasal endoscopy enables one to obtain data for culture-directed antibiotic therapy. Treatment is usually medical, in the absence of orbital or intracranial complications.

◆ **Epidemiology**

True incidence is difficult to establish because of some overlap with other complaints, such as upper respiratory infection or allergy. However, rhinosinusitis is among the most common of all healthcare complaints. Rhinosinusitis (all varieties) is reported to affect 31 million people in the United States. An estimated 20 million cases of acute bacterial rhinosinusitis occur annually in the United States. Annual U.S. expenditure related to the primary diagnosis of sinusitis totals approximately \$3.5 billion.

◆ **Clinical**

Signs and Symptoms

Acute rhinosinusitis may be suspected based on signs and symptoms. Major factors include facial pain or pressure, congestion or fullness, nasal obstruction, nasal discharge, purulence, or discolored postnasal drainage, hyposmia/anosmia, purulence in nasal cavity on exam, and fever. Minor factors include headache, fatigue, halitosis, dental pain, cough, and ear pain or ear pressure/fullness.

Updated guidelines have simplified diagnostic criteria: purulent drainage with nasal obstruction and facial pain/fullness/pressure persisting 10 days, or worsening within 10 days after initial improvement. Fever is relatively specific to acute rhinosinusitis versus other forms of sinonasal disease. Localizing symptoms may suggest specific paranasal sinus involvement: cheek or upper dental pain with maxillary sinusitis, forehead pain, or frontal headaches with frontal sinusitis. Retro-orbital or occipital pain may be seen with sphenoid sinusitis. Pervasive sinus disease can, however, remain occult with nonspecific symptoms.

Differential Diagnosis

In the early phase of the disease (first 10 days), the etiology is presumed to be viral. Thus, a typical viral upper respiratory infection is the main alternative diagnosis in the patient with a history of symptoms of less than 2 weeks' duration. Other entities to be considered include allergic rhinitis exacerbation, unrecognized chronic rhinosinusitis, or rare nasal manifestations of systemic disease such as limited Wegener's granulomatosis or sarcoid. Other causes of localized symptoms include severe periodontal disease or recurrent migraine, which may include throbbing localized headache as well as nasal congestion. In the immunocompromised patient, a high index of suspicion for invasive fungal rhinosinusitis is critical.

◆ Evaluation

The diagnosis is generally made on the basis of signs and symptoms, by history in combination with objective exam findings. Occasionally, radiographic assessment is needed.

Physical Exam

A full head and neck examination is performed, including a cranial nerve exam. It is important to exclude evidence of complicated sinusitis, such as orbital or intracranial extension of disease. Therefore, note is made of proptosis, periorbital edema, extraocular motility, tenderness, and meningeal signs. Guidelines recommend anterior rhinoscopy or nasal endoscopy; nasal endoscopy is more informative. Assessment includes position of the septum; presence of mucosal edema; presence, location, and quality of mucus or purulence; and the presence and quality of polyps or masses. A calcium alginate swab (Calgiswab, Puritan Medical Products, Guilford, ME) or suction trap can be easily used to obtain a sample of any purulence endoscopically from the sinus ostia or middle meatus for culture and sensitivities, especially in recurrent sinusitis or previous antibiotic failure.

Imaging

The diagnosis of uncomplicated acute rhinosinusitis is generally made on history and exam. If complications or alternative diagnoses are suspected, thin-section noncontrast coronal and axial computed tomography (CT) scanning of the paranasal sinuses is the most useful study. The presence of fluid- or soft tissue-density opacification of a paranasal sinus is diagnostic. Mucosal thickening without obvious fluid suggests chronic disease. The bony detail of the skull base and orbits should be assessed. Bone erosion or

thickening, or the presence of a sinonasal mass, suggests other than acute rhinosinusitis and will prompt additional work-up.

The utility of plain X-ray films is very limited. Disease of the middle meatus, infundibulum, frontal recess, anterior ethmoidal air cells, and superior nasal cavity is not identifiable. Furthermore, the radiation exposure from a screening sinus CT (i.e., limited cuts) is probably comparable to that from a plain film series.

Labs

Unless the patient is toxic or immunosuppressed, or one suspects complications of acute rhinosinusitis (orbital or intracranial extension), blood work such as a complete blood count (CBC) with differential is not helpful. Testing for allergy and immune function is considered with recurrent acute rhinosinusitis or chronic disease.

Microbiology

It is generally accepted that viral infection predominates for the first 10 to 14 days and then leads to sinus ostia obstruction. Inspissated mucus leads to bacterial infection. The most common organisms are *Streptococcus pneumoniae*, *Hemophilus influenzae*, and *Moraxella catarrhalis* as well as anaerobes and other *Streptococcus* species. Drug resistance is a growing problem.

◆ Treatment Options

Uncomplicated acute rhinosinusitis is observed or treated medically. If symptoms persist beyond 7 to 10 days, bacterial infection is more likely and antibiotics are indicated. The typical duration of therapy is 5 to 10 days. Adult empiric therapy published guidelines recommend (a) amoxicillin/clavulanate (1.75–4.0 g/250 mg per day, or 875 mg twice daily), (b) amoxicillin (1.5–4 g/d, or 500 mg three times daily), or (c) cefuroxime axetil (250 mg twice daily). In β -lactam-allergic patients, guidelines recommend (a) trimethoprim-sulfamethoxazole double-strength (twice daily); (b) doxycycline (100 mg twice daily); or (c) macrolide therapy (**Table 4.6**). If there is no improvement in 72 hours, or if there has been antibiotic use within the previous 4 to 6 weeks, the antibiotic should be switched; consider fluoroquinolone, ceftriaxone, or clindamycin. If possible, antibiotic therapy should be culture-directed at this point.

Table 4.6 Adult empiric antibiotic therapy for acute rhinosinusitis

Nonallergic	β -lactam-allergic
Amoxicillin (1.5–4 g/d) Amoxicillin–clavulanate (1.75–4.0 g/250 mg/d) Cefuroxime axetil (250 mg twice daily)	Trimethoprim–sulfamethoxazole double-strength (twice daily) Doxycycline (100 mg twice daily) Macrolide therapy (e.g., clarithromycin 500 mg twice daily)
If ineffective, or recent antibiotic use, consider fluoroquinolone; other alternatives include clindamycin and ceftriaxone.	

Note: Culture-directed treatment is highly recommended.

In pediatric patients, guidelines recommend (a) amoxicillin/clavulanate (90 mg/6.4 mg/kg per day), (b) amoxicillin (45–90 mg/kg per day), or (c) cefuroxime axetil (generally 30 mg/kg per day divided every 12 hours; maximum 1,000 mg daily). In β -lactam-allergic patients, guidelines recommend trimethoprim-sulfamethoxazole (6–10 mg trimethoprim per kg per day divided every 12 hours), or a macrolide such as clarithromycin (15 mg/kg per day divided every 12 hours).

Additional medical therapy involves the use of oral decongestants such as pseudoephedrine (30–60 mg every 6 hours as needed). Caution should be used in patients with prostatic hypertrophy or poorly controlled hypertension. Topical decongestants such as oxymetazoline 0.05% (two sprays each nostril twice daily) can be used for 3 to 5 days at most to facilitate drainage. Prolonged use will cause rhinitis medicamentosa. Oral antihistamines are indicated only if the symptoms are associated with allergy exacerbation. Topical nasal steroids have been recommended recently to help decrease inflammation and to play a prophylactic role following resolution of symptoms in patients with recurrent disease. Nasal saline is useful for thinning secretions, as is guaifenesin (600–1,200 mg twice daily).

Occasionally, maxillary puncture is necessary to obtain material for culture and/or to relieve severe symptoms. This may be done via the inferior meatus or canine fossa. A common indication for this is an immunocompromised patient on multiple recent antibiotics with an occluded sinus.

◆ Outcome and Follow-Up

Acute rhinosinusitis generally resolves with appropriate medical treatment. Subacute rhinosinusitis progresses from 4 to 12 weeks and usually still resolves. Certain patients may relapse, termed recurrent acute rhinosinusitis, with symptoms again lasting less than 4 weeks and recurring four or more times per year. In patients with recurrent disease, CT scanning and more aggressive therapy, including systemic steroids and/or surgery, may be indicated.

Meningitis is the most common intracranial complication from acute rhinosinusitis, often from sphenoid disease. This is managed with IV antibiotics and usually surgical drainage. Orbital and intracranial complications are discussed in detail in prior chapters.

4.2.2 Chronic Rhinosinusitis

◆ Key Features

- Chronic rhinosinusitis is an inflammatory condition of the nose and sinuses.
- By definition of chronic, signs and symptoms last 3 months or longer.
- Several subtypes exist, but their exact definitions remain controversial.

Chronic rhinosinusitis (CRS) is defined as a group of disorders characterized by inflammation of the mucosa of the nose and paranasal sinuses of at least 12 consecutive weeks' duration. Acute rhinosinusitis has been defined as lasting up to 4 weeks. Subacute rhinosinusitis lasts 4 to 12 weeks. CRS occurs secondary to uncontrolled acute or subacute rhinosinusitis and many times requires surgical intervention. Underlying factors such as allergies, cilia motility disorders, and immunodeficiency disorders should be considered in refractory patients.

◆ Epidemiology

Rhinosinusitis (all varieties) is reported to affect 31 million people in United States. CRS results in an estimated 24 million physician visits annually in the United States, 90% of which result in a prescription. U.S. annual expenditure related to the primary diagnosis of sinusitis totals approximately \$3.5 billion.

◆ Clinical

Signs and Symptoms

According to updated guidelines, CRS is diagnosed by (1) ≥ 12 weeks of two or more of the following symptoms: mucopurulent drainage, congestion, facial pain/pressure/fullness, decreased olfaction; along with (2) inflammation documented by endoscopic finding of pus, edema or polyps and/or positive computed tomography (CT) scan. Validated quality-of-life (QOL) assessment tools (such as the 20-item Sino-Nasal Outcome Test [SNOT-20] by JF Piccirillo) have indicated that CRS has a large negative impact on patients.

Differential Diagnosis

Intranasal neoplasia, benign or malignant, may present similarly to inflammatory disease. Neoplasms include papillomas (inverting, cylindrical), squamous cell carcinomas or adenocarcinomas, salivary tumors, sarcomas, mucosal melanoma, schwannomas, osteomas, angiofibroma, and olfactory neuroblastoma (esthesioneuroblastoma). Other entities to be considered include nasal manifestations of systemic disease such as limited Wegener's granulomatosis, Churg-Strauss's syndrome, sarcoid, tuberculosis, leprosy, or syphilis. Other causes of localized symptoms include severe periodontal disease or recurrent migraine, which may include throbbing localized headache as well as nasal congestion (primary care providers may label cephalalgia as "sinus headaches," independent of the presence of active sinonasal disease). Temporomandibular joint (TMJ) disorders may cause "sinus" headaches. Rhinitis medicamentosa is common, and patients do not always admit to use of nasal decongestants. In the immunocompromised patient (neutropenia, uncontrolled diabetes mellitus), a high index of suspicion for invasive fungal rhinosinusitis is critical. In certain clinical situations, consider Churg-Strauss's syndrome (vasculitis, asthma, eosinophilia), eosinophilic granuloma spectrum (Langerhans cell histiocytosis), T-cell lymphoma (formerly considered midline lethal granuloma), rhinoscleroma (caused by *Klebsiella rhinoscleromatis*), or rhinosporidiosis (caused by *Rhinosporidium seeberi*; endemic in India and Sri Lanka).

Table 4.7 Exam findings in chronic rhinosinusitis

CT findings	Nasal endoscopy findings
• Mucosal thickening • Bone changes • Air-fluid levels	• Discolored nasal drainage • Nasal polyps • Edema or erythema, especially at middle meatus or ethmoidal bulla

◆ **Evaluation**

Controversy exists with regard to definitions and diagnoses of all forms of rhinosinusitis. A diagnosis is generally made on the basis of history in combination with objective exam findings and radiographic assessment, usually CT scanning (**Table 4.7**). As mentioned under Signs and Symptoms, updated 2015 evidence-based guidelines are simplified. Olfactory testing and the SNOT-20 questionnaire are strongly encouraged.

Physical Exam

A full head and neck exam is performed, including a cranial nerve exam. It is important to exclude evidence of complicated sinusitis, such as orbital or intracranial extension of disease. Therefore, note is made of proptosis, periorbital edema, extraocular motility, tenderness, and meningeal signs. Rigid nasal endoscopy should be performed after topical decongestion. Assessment includes position and integrity of the septum; presence of mucosal edema; presence, location, and quality of mucus or purulence; and the presence and quality of polyps or masses. A calcium alginate swab (Calgiswab, Puritan Medical Products, Guilford, ME) or suction trap can be easily used to obtain a sample of any purulence endoscopically from the sinus ostia or middle meatus for culture and sensitivities. The otolaryngologist possesses unique expertise in nasal endoscopy, and this procedure should be part of the exam of all patients presenting with sinonasal complaints.

Consider olfactory testing (especially in patients who note hyposmia/anosmia as a complaint), such as the Smell Identification Test (Sensonics, Inc., Haddon Heights, NJ), formerly called the University of Pennsylvania Smell Identification Test (UPSIT).

Imaging

Thin-section noncontrast coronal and axial CT scanning of the paranasal sinuses is the most useful study. CT staging systems (i.e., Lund-Mackay) have been proposed and may be useful for research or tracking disease over time. CT findings correlate poorly with patients' symptoms. The presence of fluid or soft tissue density opacification of a paranasal sinus is diagnostic. Mucosal thickening without obvious fluid suggests chronic disease. The bony detail of the skull base and orbits should be assessed. Bone erosion or thickening, or the presence of a sinonasal mass, suggests other than acute (chronic) rhinosinusitis and will prompt additional work-up. Fungal disease, presenting as allergic fungal rhinosinusitis, will appear as heterogeneous density. Evidence of expansile disease with bone thinning is seen with

allergic fungal rhinosinusitis, mucocele, and low-grade neoplasms. Bony erosion should raise concern for malignancy and is also seen with inflammatory disease such as Wegener's granulomatosis. Bony thickening is seen with long-standing inflammatory disease.

Magnetic resonance imaging (MRI) is helpful for assessing intracranial extension. Fungus may appear as absent signal with both T1 and T2 sequences. However, at present MRI is not recommended as an alternative to CT for routine diagnosis of chronic rhinosinusitis because of its excessively high sensitivity and lack of specificity. The utility of plain X-ray films is very limited.

Labs

Unless the patient is toxic or immunosuppressed, or one suspects complications of rhinosinusitis (orbital or intracranial extension), blood work such as a CBC with differential is not helpful. Bacterial cultures from sinus ostia are useful. The role of fungi is controversial. Fungi are ubiquitous; topical or systemic antifungals are not used unless tissue-invasive fungal infection is present (i.e., in immunosuppressed patients). The role of virus in chronic rhinosinusitis is unclear and is under study.

Microbiology

Inspissated mucus leads to bacterial infection. Importantly, organisms differ from those seen with acute rhinosinusitis. Common microbes include *Staphylococcus aureus*, coagulase-negative *Staphylococcus*, and anaerobic and gram-negative species. Most infections are polymicrobial. Antimicrobial resistance is common.

Pathogenesis and Classification

Etiology is multifactorial, involving environmental, local host, and general host factors. Pathogenic environmental factors include viral infection, air pollution, smoking, and allergy. Smoking is the single most significant factor in recurrent disease; tobacco cessation is strongly encouraged before sinus surgery is performed. Host factors include atopy, immune deficiency, mucociliary clearance problems such as ciliary dyskinesia or cystic fibrosis (probably even subclinical forms of chloride transporter dysfunction), airway hyperactivity, and reactivity to fungus.

Subtypes of chronic rhinosinusitis have been proposed based on inflammatory infiltrate (eosinophilic, neutrophilic, or other) or on various pathophysiologic mechanisms (extrinsic factors, intrinsic factors). The clinical utility of classification has been hampered to some degree by overlap in signs and symptoms. However, categories include chronic rhinosinusitis with or without polyps, chronic rhinosinusitis with eosinophilic mucin, allergic fungal rhinosinusitis (based on the presence of eosinophils and atopy to fungus), and Samter triad (ASA triad) of allergy to aspirin, with asthma, and nasal polyps. CRS is extremely common in patients with acquired immunodeficiency syndrome (AIDS). Finally, the microbiome is an area of active study, although standard measure techniques or treatment recommendations based on microbiome measures are not yet available.

Table 4.8 Treatment strategies in chronic rhinosinusitis

Oral steroids (consider obtaining consent)	Methylprednisolone 32 mg daily × 7 days, then taper
Oral antibiotics	Culture-directed whenever possible
Topical nasal steroid sprays	Mometasone, fluticasone, other
Topical nasal steroid irrigation (off-label use)	Budesonide (one Respule contents added to 240 mL nasal saline) twice daily
Topical nasal antibiotic irrigation, useful for methicillin-resistant <i>S. aureus</i> (MRSA; off-label use)	2% mupirocin ointment 22 g in 1 L of normal saline using 50 mL per nostril daily twice daily for 10 days
Surgery	Endoscopic (forceps, microdebrider, balloon catheter dilation)
	Open (trephination, frontal obliteration, other)

◆ **Treatment Options**

Medical

Uncomplicated chronic rhinosinusitis should be treated medically prior to considering surgery (**Table 4.8**). There is debate regarding what constitutes “maximal medical therapy.” Treatment is aimed at infection, inflammation, and underlying factors. A course of systemic steroid tapered over 1 month combined with oral antibiotic therapy has been found useful, especially if polyps are present. Note that oral steroids have risk of serious side effects, and for off-label use it is highly recommended that consent be documented. Methylprednisolone (Medrol, Pfizer, New York, NY) 32 mg orally every day for 1 week, then tapered to 32 mg every other day for 1 week, 24 mg every other day for 1 week, 16 mg every other day for 1 week is a typical regimen. Whenever possible, antibiotic therapy should be culture-directed. Empiric choices include cefuroxime 250 to 500 mg orally twice a day; clarithromycin and doxycycline are alternatives. Antibiotic treatment for CRS may be prolonged. At the end of this treatment, a CT scan is repeated (with image-guidance protocol, if surgery is likely). If symptoms and/or exam and CT findings are not significantly improved, surgery is considered.

Nasal saline irrigations and topical nasal steroids are recommended in updated guidelines. Additional useful medical therapy may include guaifenesin 600–1200 mg orally twice a day (< 1 week) and saline nasal rinses. Methicillin-resistant *S. aureus* (MRSA) cultured from nasal endoscopy specimens has been seen at initial otolaryngology visit. Nasal irrigation with mupirocin ointment 22 g in 1 L of normal saline using 50 mL per nostril daily for 10 days appears beneficial (this is off-label use). Topical decongestants such as oxymetazoline 0.05% (two sprays each nostril twice daily) can be used for 3 to 5 days at most to facilitate drainage. Prolonged use will cause rhinitis medicamentosa. Oral antihistamines are indicated only if the symptoms are associated with allergy exacerbation. Budesonide (Pulmicort

Respules, AstraZeneca, Cambridge, UK; one Respule contents added to 240 mL nasal saline, twice daily for topical nasal irrigation) has been used in some patients with CRS with polyps as a systemic steroid-sparing approach (this is off-label use).

Surgical

Endoscopic sinus surgery has become the standard technique for addressing nonneoplastic sinonasal disease. It is also being used for selective neoplastic cases. Open techniques may be necessary in certain cases; for example, open drainage of orbital subperiosteal abscess, or frontal sinus trephination in combination with endoscopic surgery. Preoperative review of the coronal CT scan is critical (Table 4.9). Pre- and postoperative care is important for success. Many surgeons employ pretreatment, especially in polyp cases, with methylprednisolone 32 mg orally daily for 5 days prior to surgery, tapered slowly postoperatively, depending on endoscopic exam. Tobacco cessation is mandatory to long-term success. The role and extent of office postoperative débridement is debated.

Endoscopic surgical instrumentation may include through-cutting forceps or powered microdébriders. Tissue-sparing instruments for balloon sinus dilation over a guidewire are widely used.

◆ Outcome and Follow-Up

Chronic rhinosinusitis patients require long-term ongoing care, often including the allergist, rhinologist, and primary physician. Cumulative systemic steroid exposure must be carefully followed to prevent side effects. There is a high rate of revision surgery in CRS patients, especially those with extensive nasal polyposis. It is important for patients to understand that they have a chronic disease condition involving the upper airway and that curing this chronic condition may not be achievable. However, effectively managing symptoms, minimizing disease morbidity, and improving quality of life are reasonable goals. Serial SNOT-20 QOL assessments are useful. Orbital and intracranial complications are discussed in detail elsewhere in this book.

Table 4.9 Preoperative review of coronal CT scan

Shape and integrity of skull base	Height of ethmoidal air cells (in relation to top of maxillary sinus posteriorly—number of cells before skull base)
Shape and integrity of orbit	Sphenoidal sinus anatomy: septations, carotids, optic nerves
Position of uncinate	Anatomic variations: Onodi cell (overriding posterior ethmoidal air cell) Concha bullosa Infraorbital ethmoidal air cell
Location of anterior ethmoidal artery	Hypoplastic maxillary sinus

4.3 Rhinitis

4.3.1 Nonallergic Rhinitis

◆ Key Features

- Many physicians fail to recognize nonallergic rhinitis (NAR), instead treating all rhinitis as allergic.
- Even when allergy has been ruled out, these patients may be diagnosed with “vasomotor rhinitis” (VMR), often a wastebasket diagnosis. Further work-up is often not performed.
- Because NAR encompasses a variety of etiologic factors, careful diagnosis and directed therapy are key to satisfactory treatment.

Rhinitis is defined as inflammation of the nasal lining. It is characterized by symptoms of nasal congestion, rhinorrhea, sneezing, and/or nasal itching. Rhinitis is broadly classified into allergic (AR) and nonallergic (NAR) categories, based on skin testing or in vitro tests for allergen-specific immunoglobulin E (IgE). The term “mixed rhinitis” refers to the presence of both allergic and nonallergic components, and treatment should address both components.

◆ Epidemiology

In the United States, NAR affects 19 million people, and an additional 26 million experience mixed rhinitis. NAR disproportionately affects females, suggesting hormonal influences. It more commonly affects people older than 60 years of age.

◆ Clinical

Signs and Symptoms

Often, VMR is associated with nasal obstruction and thickened postnasal drip. In addition, it can present with copious, watery anterior rhinorrhea, often triggered by changes in temperature, alcohol use, or exposure to odors and aromas. Other common symptoms of NAR include sneezing, congestion, and itching, which can be triggered by inhaled agents, cigarette smoke, foods, chemicals, and medications. All can have an irritant effect on the nasal mucosa.

Differential Diagnosis

NAR is diagnosed after AR has been excluded. Stimulation of the nasal mucosa by endogenous and exogenous agents causes symptoms that characterize rhinitis. In addition, NAR can be exacerbated by hormonal influences such as those experienced during pregnancy, endocrine disturbances such as hypothyroidism, and autonomic dysfunction.

◆ Classification

NAR can be classified into the following subtypes: occupational (irritative-toxic), hormonal, drug-induced, gustatory, inflammatory (eosinophilic), and vasomotor. Rhinitis due to vasculitides and granulomatous diseases (Wegener, sarcoidosis, etc.) is a separate category.

Occupational Rhinitis

Occupational (irritative-toxic) rhinitis is caused by inhalant irritants or toxic agents including chemicals, solvents, and cigarette smoke. Cigarette smoking is the most common cause, and occupational exposure the next leading cause, of this subtype. Cessation and adequate protection from these stimuli result in control of symptoms.

Hormonal Rhinitis

The most commonly reported hormonal rhinitis is rhinitis of pregnancy, seen in 22% of nonsmoking women and in 69% of smokers. Rhinitis of pregnancy is most common during the later stages of pregnancy. Rhinitis of pregnancy resolves within 2 to 4 weeks of delivery.

Drug-Induced Rhinitis

Many common medications have physiologic side effects on the nose, resulting in nasal congestion. These medications include angiotensin-converting enzyme (ACE) inhibitors, β -blockers, oral contraceptives, antipsychotics, and recently, phosphodiesterase type-5 inhibitors. Some patients may have a non-IgE-mediated sensitivity to nonsteroidal anti-inflammatory drugs (NSAIDs) and aspirin.

Rhinitis medicamentosa is most commonly associated with prolonged use of topical vasoconstrictive nasal sprays such as oxymetazoline. Prolonged use of these medications results in tachyphylaxis, and cessation of use is associated with rebound nasal congestion, which can be severe and refractory to treatment.

Gustatory Rhinitis

Gustatory rhinitis is characterized by watery, often profuse rhinorrhea that develops with ingestion of food. Anticipation of eating or even the smell of food may precipitate clear nasal drainage. The response is mediated by aberrant parasympathetic stimulation of nasal secretomotor fibers on stimulation of salivary secretomotor activity.

Nonallergic Rhinitis with Eosinophilia

Nonallergic rhinitis with eosinophilia (NARES) was originally described as a constellation of perennial sneezing attacks, profuse watery rhinorrhea, and nasal pruritus, as well as nasal congestion in patients who showed a lack of evidence for allergy on skin testing or in vitro testing for specific IgE and who had greater than 20% eosinophils on nasal smears.

Vasomotor Rhinitis

VMR is the most common form of chronic NAR. VMR is a diagnosis of exclusion. VMR is widespread, especially in the elderly, and may be frustrating to treat.

Chemical Sensitivity

A particularly severe form of sensitivity to certain chemical substances is seen on occasion. History generally describes an exposure to a strong or caustic substance with onset of severe or disabling rhinitis symptoms on repeat exposure to even mild or normal chemicals, such as perfumes or cleaning agents. This is poorly understood and has been considered a “multiple chemical sensitivity syndrome.” Strict avoidance, if possible, may be effective, although better treatment options remain to be determined.

◆ Evaluation

History

Family and personal history of atopy or allergies, medications, abnormal hormonal status, pregnancy, smoking, and occupational exposure are noted.

Physical Exam

The diagnosis of rhinitis involves a complete head and neck examination, as well as a nasal endoscopy. The size of the nasal conchae and the degree of nasal airway impairment are noted. The reversibility of this hypertrophy is gauged after spraying with topical decongestants. The appearance of the nasal mucosa and secretions is evaluated. Allergic mucosa is classically blue-gray and boggy, whereas erythematous, inflamed mucosa is seen in rhinitis medicamentosa, irritative rhinitis, and granulomatous rhinitis.

Labs

AR should be ruled out by skin allergy testing or by in vitro tests quantifying specific IgE levels. NAR eosinophilic sinusitis can be ruled out by nasal wash cytology or by nasal brushings for eosinophils.

Other Tests

Anterior rhinomanometry, acoustic rhinometry, and nasal peak flows are supplementary tests more commonly used as research tools and not routinely used for diagnosis. Other available tests include serum mediator levels, tests for complement activation, and nasal lavage studies. Nasal lavage studies of the total protein and albumin concentration show significantly higher levels in AR patients as compared with NAR patients and healthy control subjects, likely due to increased vascular permeability.

◆ Treatment Options

Medical

Conservative management should be attempted first. This includes avoidance of irritating substances, cold weather, offending foods, and wine. Regular physical exercise also increases sympathetic input to the nose, correcting

for the underlying loss of sympathetic tone. Underlying diseases and medications should be evaluated. Topical anticholinergics, antihistamines, and steroids are commonly used to treat NAR.

Azelastine is approved for use in both AR and NAR. Azelastine is an H₁-receptor antagonist. It also inhibits synthesis of leukotrienes, kinins, and cytokines. It also suppresses intercellular adhesion molecules expression and superoxide free radical generation. This anti-inflammatory effect provides relief in NARES and VMR. The antihistaminic effect decreases mucosal edema, prostaglandin production, and stimulation of irritant receptors.

Anticholinergic medications provide relief from rhinorrhea in NAR. Ipratropium bromide is a topical anticholinergic with uncommon systemic side effects. Nasal steroid sprays can also control nasal congestion and rhinorrhea effectively.

Vasculitides and autoimmune granulomatous rhinitis (Wegener's granulomatosis, sarcoidosis, Churg-Strauss's disease) are treated with immunosuppressive drugs.

Surgical

The inferior conchae are lined with vascular mucosa with mucous and serous glands and also contain venous sinusoids surrounded by smooth muscle fibers under autonomic control. Inferior conchal hypertrophy unresponsive to medical therapy can be treated surgically with conchal reduction by a variety of techniques.

Vidian neurectomy and sphenopalatine ganglion block have also been used for intractable VMR. Botulinum toxin (Botox, Pfizer, New York, NY) injection to the sphenopalatine ganglion has also been recently described.

◆ Outcome and Follow-Up

If left untreated, patients continue to suffer from detrimental effects on quality of life. Lifestyle modifications and pharmacologic interventions can be useful, but some forms of NAR, especially in the elderly, are particularly challenging to manage. Once symptoms are adequately controlled, patients can be managed with semiannual or annual follow-up.

4.3.2 Allergy

◆ Key Features

- An allergy is an immune response with deleterious effects.
- Emergencies include anaphylactic shock and angioedema.
- Signs and symptoms often involve otorhinolaryngologic manifestations.

The immune system functions to protect a host from foreign antigens. An allergic reaction is an immune response that causes undesirable effects.

Examples of allergic disease of importance to the otolaryngologist are allergic rhinitis, angioedema, latex allergy, and anaphylaxis.

◆ **Epidemiology**

An estimated 17% of the population is affected by allergic disease.

◆ **Clinical**

Important to the clinical management of allergic disease is an understanding of the four types of allergic reactions, as described by Gell and Coombs (**Table 4.10**). Type I is an immediate immunoglobulin E (IgE)-mediated reaction (e.g., urticaria or anaphylaxis); type II is an IgG- or IgM-mediated cytotoxic reaction (e.g., transfusion reaction); type III is a response due to immune complexes, usually IgG (e.g., glomerulonephritis); type IV is a delayed hypersensitivity reaction mediated by T cells (e.g., poison ivy rash, granuloma formation). Inflammatory mediators such as histamine, leukotrienes, and cytokines mediate the effects of an allergic reaction.

Signs and Symptoms

Signs and symptoms may vary widely depending on the allergen and exposure. Nasal effects include sneezing, rhinorrhea, itching, obstructive edema, and supratip crease with so-called allergic salute. Ophthalmologic effects include conjunctival injection, itching, elongated lashes, and allergic shiners. Other airway effects may include chronic mouth breathing, globus sensation with frequent throat clearing, cobblestone appearance to the posterior pharynx, coughing, and wheezing.

Acute allergic emergencies include anaphylaxis and angioedema. Anaphylaxis may present with rapid onset of bronchospasm, laryngeal edema, cough, stridor, itching, urticaria, tachycardia, nausea, and initial hypertension followed by hypotension and cardiovascular collapse. Death may occur rapidly due to airway obstruction or cardiovascular collapse (shock). Angioedema may present with swelling of the lips, oral cavity, tongue, and/or larynx. Progression may be unpredictable and rapid, causing death secondary to airway obstruction.

Table 4.10 Gell and Coombs classification of allergic reactions

Type	Reaction
I: Immediate IgE response, mast cell degranulation	Response within minutes; may include wheezing, urticaria, rhinorrhea, angioedema, anaphylaxis
II: Cytotoxic IgG/IgM reaction	Examples: transfusion reaction with hemolysis; hyperacute graft rejection
III: Immune complex reaction, IgG	Often delayed; may affect various tissues Example: glomerulonephritis, arthritis
IV: Cell-mediated, T-cell response	Example: poison ivy, granuloma formation

Abbreviation: Ig, immunoglobulin.

Differential Diagnosis

Patients presenting with complaints or findings related to possible allergic etiologies may be complex. Considerations include inhalant allergy, food allergy, side effects of medications such as ACE inhibitors, immune disorders such as C1 esterase inhibitor deficiency, various immune deficiencies, or infection.

◆ Evaluation

Work-up must be tailored to the urgency of the situation. In an emergency, evaluation may be limited to a primary assessment, vital signs, and an assessment of the upper airway followed by immediate treatment directed at angioedema or anaphylactic shock. The elective setting enables one to perform a thorough history and physical examination along with appropriate ancillary testing.

Physical Exam

A full head and neck exam is standard. Auscultation of the lungs for wheezing is performed. Note evidence of chronic allergy, such as allergic shiners, supratip nasal crease, as well as nasal conchal edema, nasal mucus, postnasal drainage, and posterior pharyngeal cobblestoning. As for laboratory testing, a scraping of the inferior conchal mucosa for cytology may suggest allergic disease—presence of eosinophils, or nasal mastocytosis. Consider C1 esterase inhibitor levels in cases of recurrent angioedema.

Other Tests

Specific allergy testing may be performed either via blood sample for radioallergosorbent test (RAST) or via skin testing. RAST is useful as a screening for allergen-specific IgE but is insufficient to begin immunotherapy. Skin testing, to elicit a wheal and flare response to specific allergens, can be performed either via the prick test or the serial dilution end point titration (SDET) method, endorsed by the American Academy of Otolaryngic Allergy. SDET provides quantitative data useful for safely beginning immunotherapy. Some allergists use the results of RAST to prescribe immunotherapy. Details of skin testing are beyond the scope of this book; instead see Marple BF, Mabry RL. *Quantitative Skin Testing for Allergy: IDT and MQT*. New York, NY: Thieme; 2006.

◆ Treatment Options

Anaphylaxis

Immediate treatment for anaphylaxis includes airway and circulatory support. Airway support with supplemental oxygen, intubation, or a surgical airway may be indicated. Epinephrine 0.3 mg intramuscularly (IM) or subcutaneously (SC) repeated every 10 minutes up to 1 mg is given (adult dose). Two large-bore intravenous (IV) cannulas should be placed. IV crystalloid bolus, IV diphenhydramine 50 mg, IV H₂ blocker, and IV dexamethasone 8 mg are administered. ICU observation is planned, with further circulatory management with dopamine drip as indicated.

Angioedema

Airway obstruction must be anticipated. IV access and continuous airway monitoring are mandatory. IV diphenhydramine 50 mg, IV H_2 blocker, and IV dexamethasone 8 mg are administered. If airway obstruction is significant, if swelling persists, or if swelling worsens despite medical treatment, elective intubation is performed via awake fiberoptic nasotracheal technique, with a tracheotomy tray available. Direct laryngoscopy is contraindicated, as sedation and paralysis may lead to an inability to mask ventilate, and tongue edema will interfere with visualization of the glottis.

Chronic Allergy Management

Pharmacotherapy includes H_1 blockers, topical nasal steroids, oral steroid tapers used sparingly, topical nasal cromolyn, leukotriene inhibitors, oral decongestants, and topical decongestants used for a maximum of 3 consecutive days to avoid rhinitis medicamentosa. Immunotherapy injections for desensitization may be indicated, based on results of skin testing. Sublingual immunotherapy is an option that has been gaining acceptance, although at present insurance coverage has been less than satisfactory because it has been considered “off-label use.” Environmental measures are critical, in terms of avoidance of exposure to allergens such as dust mites, molds, or pets.

◆ Outcome and Follow-Up

Routine follow-up for response to pharmacotherapy and for immunotherapy is indicated. Consultation with a pulmonologist for management of lower airway disease may be helpful.

4.4 Inverted Papillomas

◆ Key Features

- An inverted papilloma is a soft tissue neoplasm often arising from the lateral nasal wall.
- It is benign but can be locally aggressive.
- Malignant degeneration can occur.

An inverted papilloma is considered a benign neoplasm. However, reports suggest that as many as 10 to 20% of cases can contain, or degenerate to, in situ or invasive squamous cell carcinoma. Thus, complete surgical removal is the treatment of choice.

◆ Epidemiology

Males are affected three times more frequently than females. The tumor is more common in older adults.

◆ Clinical

Signs

The patient presents with a polypoid unilateral intranasal mass. Any patient presenting with a unilateral nasal mass should raise suspicion for neoplasm. Allergic fungal sinusitis may present unilaterally as well.

Symptoms

The patient has a unilateral nasal obstruction, with or without sinusitis. Rhinorrhea and/or epistaxis may occur. Histologically, an inverted papilloma consists of in-folded epithelium that may be squamous, transitional, or respiratory.

Differential Diagnosis

The differential diagnosis includes an inverted papilloma, fungiform papilloma (often arises from anterior nasal septum), cylindrical papilloma (often arises from lateral wall; rare tumor), minor salivary gland benign or malignant tumors, lacrimal sac tumors, olfactory neuroblastoma, carcinoma such as squamous cell carcinoma or sinonasal undifferentiated carcinoma, mucosal melanoma, chondrosarcoma, angiofibroma, inflammatory nasal polyp, allergic fungal rhinosinusitis. Other tumors include schwannomas, hamartomas, giant cell granulomas, neurofibromas, and chondromyxoid fibromas.

◆ Evaluation

History

A standard history is taken.

Physical Exam

A full head and neck exam is done, with attention paid to the cranial nerves. Nasal endoscopy with a rigid 0° or 30° endoscope will be useful to assess the intranasal extent and location of soft tissue mass. After imaging excludes possible encephalocele, a biopsy may be obtained in the office or operating room.

Imaging

Computed tomography (CT) provides bony detail and is reviewed with attention to erosive changes at the orbit and skull base. Magnetic resonance imaging (MRI) is helpful in assessing soft tissue tumor extent and can distinguish between inspissated secretion and tumor. CT will often overestimate tumor extent. Images should be reviewed in axial, coronal, and sagittal planes.

◆ Treatment Options

Medical

Definitive treatment requires surgery. Radiotherapy is ineffective and may in fact induce carcinoma.

Surgical

Complete tumor excision is required. Inadequate removal can lead to recurrence, and malignant transformation can occur. Therefore, as summarized by Myers et al, the surgical approach for the management of this tumor must allow (1) adequate exposure for complete removal, (2) an adequate view of the cavity for postoperative examination, and (3) acceptable cosmetic and functional results. Two approaches are commonly employed at present: the open approach via lateral rhinotomy for medial maxillectomy, and the endoscopic approach. Although there has been ongoing debate, most surgeons agree that the procedure utilized should be individualized to the size and location of the tumor and that endoscopic removal can be done in a manner that fulfills these criteria, with a recurrence rate no higher than that for an open approach, in appropriate cases. A midface degloving approach can also be used, especially for tumors located inferiorly (i.e., the area of the premaxilla, nasal septum, or inferior nasal concha).

Regardless of approach, intraoperative frozen pathology should be used to ensure that margins are negative for evidence of residual tumor. Orbital or intracranial extension may require assistance of ophthalmologic or neurosurgical colleagues.

Preoperative Planning

Imaging is reviewed. CT provides bony detail and is reviewed with attention to erosive changes at the orbit and skull base. MRI is helpful for assessing soft tissue tumor extent and can distinguish between inspissated secretion and tumor. CT will often overestimate tumor extent.

A biopsy is required prior to surgery, after imaging is reviewed. Do not biopsy a possible encephalocele; review the MRI. Many surgeons perform biopsy in the operating room combined with endoscopic assessment of the tumor, given concerns for possible hemorrhage following thorough biopsy. If the biopsy indicates malignancy, an open approach to wide resection is often employed if surgery is to be done; radiation or chemoradiation may be options, depending on histology. Endoscopic skull base cancer resection may be performed if appropriate expertise is available.

Open Medial Maxillectomy

A temporary tarsorrhaphy stitch with 5-0 nylon is used to protect the eye. A lateral rhinotomy incision is performed, from just above the medial canthus, along the nasal facial groove, and around the ala, and, if needed, the lip may be split. The periosteum is elevated, as is the periorbital. The anterior and posterior ethmoidal arteries are left as skull base landmarks, indicating the superior extent of dissection. Anterior antrostomy is performed, avoiding injury to the infraorbital nerve. Osteotomies are made along the nasal bone, along the floor of the nose, below the frontoethmoid suture, and at the junction of the lamina papyracea and orbital floor. Posterior attachments are divided with heavy scissors, removing the lateral nasal wall tissue block. Internal maxillary artery branches may require control for hemostasis. Mucosa is stripped from the maxillary, ethmoidal, and sphenoidal sinuses, which are opened. The lacrimal sac is opened and sutured to surrounding

tissue. The cavity is packed with Gelfoam (Pfizer, Inc., New York, NY) and antibiotic gauze, and the wound is closed.

Endoscopic Tumor Removal

The following conditions are necessary for endoscopic tumor removal: known histology, adequate imaging studies, surgeon training/experience, and adequate instrumentation. The informed consent must discuss possible conversion to external and/or transoral open procedure. The use of intraoperative computer-assisted surgical navigation (image guidance) is often helpful for endoscopic tumor resection. The techniques used in standard endoscopic sinus surgery are employed, with the goal of complete tumor removal along with a margin of healthy tissue. The nose is topically decongested; an initial rigid endoscopic exam is done for tumor assessment. Local injections are performed using 1% lidocaine with 1:100,000 epinephrine into the sphenopalatine region, middle conchal insertion, and into tumor. Resection is performed with through-cutting instruments; the powered microdébrider is often helpful. It is possible (and typical) to remove the tumor in numerous pieces without compromising the surgery. Often, a tumor stalk can be followed to a precise site of attachment. The bone underlying this attachment site should be abraded with a diamond drill if possible. If the attachment involves a concha, a portion of the concha can be resected. Similarly, the medial wall of the maxillary sinus can be widely removed. If needed, an endoscopic medial maxillectomy may be performed. Bone cannot be examined by frozen section, although mucosa can be sent. If lateral maxillary sinus tumor extension cannot be reached endoscopically, a Caldwell-Luc anterior antrostomy can be used. Steps are taken to ensure that adequate sinus drainage pathways are present. Standard packing, antibiotics, and routine endoscopic sinus surgery postoperative care are used.

An advantage of endoscopic removal is that the exact tumor attachment site is identified and can be followed endoscopically in the office for surveillance for tumor recurrence.

◆ **Complications**

Major surgical complications can include orbital injury, optic nerve injury, skull base injury with cerebrospinal fluid (CSF) leak, and hemorrhage.

◆ **Outcome and Follow-Up**

An antibiotic such as cefuroxime axetil 250 mg orally twice a day or clindamycin 300 mg orally three times a day is used until any packing is removed, usually at postoperative day 3 or 4. Narcotic analgesic such as oxycodone-acetaminophen 5 mg/325 mg, 1 to 2 tabs orally every 6 hours as needed, is prescribed. Patients are seen in 1 week and then 3 weeks for gentle cavity débridement. Nasal saline irrigation (i.e., NeilMed Sinus Rinse) is done twice daily. Patients are typically seen every 3 to 6 months for tumor recurrence surveillance. Overall recurrence rates up to 20% have been reported, although recurrence should not be expected with adequate margins.

4.5 Anosmia and Other Olfactory Disorders

◆ Key Features

- Olfactory sensory neurons situated in the nasal cavity constitute the first cranial nerve (CN I) and relay olfactory stimuli to the olfactory bulbs of the brain.
- Anosmia, or olfactory loss, can be considered to be conductive (secondary to a process causing nasal obstruction), sensorineural (secondary to a process affecting olfactory neurons or central pathways), or mixed.

The most common causes of olfactory sensory loss are rhinosinusitis, head trauma, or postviral anosmia. Presbyosmia, or age-related decline in olfactory function, is well documented in patients over age 65. Less commonly, effects of drugs or systemic illness have been reported to impact olfactory function adversely.

Anosmia is an absence of olfactory function. Hyposmia (or microsmia) is a decrease in olfactory function. Parosmia is an altered olfactory perception in the presence of stimulus, usually considered foul. Phantosmia is an olfactory perception in the absence of stimulus.

◆ Epidemiology

Approximately 2 to 3 million Americans suffer from chemosensory dysfunction. Olfactory loss is present in 1% of those under age 60 but in > 50% of those over 60 years of age. Presbyosmia is present in most of those > 80 years of age.

◆ Clinical

Signs

In cases of conductive anosmia, nasal endoscopy will usually reveal evidence of obstructive disease, such as mucosal edema, inflammation, mucopus, nasal polyps, or other intranasal mass. In sensorineural anosmia, there may be no obvious exam findings. However, intracranial lesions often cause additional neurologic defects such as altered mental status, urinary incontinence, or seizures. Neoplasms such as an olfactory groove meningioma may cause Foster-Kennedy's syndrome (ipsilateral anosmia, optic atrophy and central scotoma, contralateral papilledema). Objective olfactory testing will reveal the sensory loss.

Symptoms

Many patients are unaware of significant olfactory loss. Those who rely on olfaction, such as cooks, perfumers, firefighters, or chemical workers, are very bothered by hyposmia or anosmia. Acute loss may follow coup/contrecoup injury from even minor head trauma. Fluctuating loss is common with chronic rhinosinusitis. Many anosmics mistakenly complain of loss of taste.

Differential Diagnosis

Obstructive intranasal lesions include polyps, infectious or inflammatory disease, and neoplasia, including Schneiderian papilloma, carcinoma, olfactory neuroblastoma, mucosal melanoma, and metastases. In the absence of obstruction, a sensorineural loss is most often posttraumatic or postviral; thus the history should be corroborative. One should also consider intracranial lesions such as meningioma, neurologic disease such as Parkinson's or Alzheimer's disease, medication side effects, or toxic exposure. Iatrogenic causes such as previous nasal or sinus surgery or prior radiation therapy should be considered. Congenital causes of anosmia include Kallmann's syndrome (hypogonadotropic hypogonadism with anosmia); if suspected, these patients should have an endocrinology consultation. Smoking has also been associated with decreased olfaction, as has the use of over-the-counter intranasal zinc-containing preparations. Other genetic causes are seen rarely, such as ciliopathies.

◆ Evaluation

Objective Olfactory Testing

Validated and easily administered tests of olfactory discrimination are readily available, such as the Smell Identification Test (Sensonics, Inc., Haddon Heights, NJ). This is a "scratch-and-sniff" multiple choice 40-item test that detects anosmia, hyposmia, and malingering and provides scores with age-adjusted norms. More sophisticated tests of olfactory threshold for specific odorants are less commonly employed outside of research centers.

Imaging

In patients without an obvious conductive etiology or history for postviral or posttraumatic anosmia, a head computed tomography (CT) or magnetic resonance imaging (MRI) scan is recommended to rule out an occult skull base or intracranial neoplasm. Thin-section coronal CT with bone algorithms provides excellent detail of the frontoethmoidal and cribriform plate region. Bone dehiscence or soft tissue mass suggests a lesion (e.g., encephalocele, olfactory neuroblastoma). MRI is the test of choice for intracranial lesions.

◆ Treatment Options

Conductive Anosmia

Medical

Medical management for infectious or inflammatory sinonasal disease includes topical nasal steroids, systemic steroid tapers, and oral antibiotics. Oral steroids in combination with culture-directed antibiotics for 4 weeks are considered by many to constitute maximal medical therapy.

Surgical

Endoscopic sinus surgery, with or without septoplasty, can be effective for relieving obstructing disease such as polyposis. Endoscopic or craniofacial approaches to sinonasal or anterior skull base tumors may be indicated.

Sensorineural Anosmia

At present, there is an absence of effective treatment options for postviral anosmia, posttraumatic anosmia, or presbyosmia. Many cases of postviral anosmia or posttraumatic anosmia may recover over 1 to 2 years, but this is difficult to predict. Many patients report parosmias early during recovery. Empiric trial of steroids is employed by many clinicians, and steroid-responsive anosmia is well documented. However, this must be weighed against the risks of steroid use. Oral zinc therapy is often recommended, but the available evidence does not support its use. Olfactory training therapy is safe and has been found to be helpful in some studies: patients are instructed to sniff three different odors (such as citrus, floral perfume, and coffee grounds) for 10 minutes twice daily for several weeks. Appropriate referral for suspected neurologic or systemic disease is imperative.

◆ Outcome and Follow-Up

It is critically important to educate the anosmic patient regarding the importance of working smoke detectors, natural gas detectors, and the labeling and dating of foods to avoid serious injury or death.

4.6 Taste Disorders

◆ Key Features

- The sense of taste is mediated by cranial nerves (CNs) VII, IX, and X.
- Most complaints of taste dysfunction are in fact olfactory disorders.
- Medication effects are the most commonly identifiable cause of taste dysfunction.

Taste or gustation is a proximal chemical sense. Unlike in olfaction (see **Chapter 4.5**), for taste the primary source of stimulus must be in physical contact with the patient. The qualitative descriptors elicited by stimulation of taste receptors include sweet, sour, salty, bitter, and umami (loanword from Japanese that roughly equates to “savory,” with its own specific receptors stimulated by glutamates).

◆ Anatomy

Epithelial taste receptor cells are located on the anterior and posterior tongue, the soft palate, and in the larynx. Taste receptor cells are arrayed in sensory end organs, the taste buds, and synapse on primary sensory neurons of the facial nerve (CN VII), the glossopharyngeal nerve (CN IX), or the vagus nerve (CN X). Taste innervation is therefore redundant; that is, a lesion of a single cranial nerve would not abolish all taste input. Taste

receptor cells in taste buds turn over and are replaced by proliferative basal cells. This indicates that certain clinical conditions that damage or destroy taste receptors may cause only temporary dysfunction since the system has a capacity for repair.

◆ Epidemiology

Taste loss, whether total (ageusia) or partial (hypogeusia), is much less common than olfactory loss. In a study of a large series of patients with chemosensory complaints, only 4% were found by objective testing to have a true taste deficit; the remainder exhibited olfactory deficits. This reflects the fact that the experience of flavor is mediated by the synergistic stimulation of the olfactory system, the oropharyngeal somatosensory system, and the actual taste receptors of the gustatory system.

◆ Clinical

Taste dysfunction may be a loss or a phantom or altered perception (dysgeusia). Partial loss is most common, and losses are usually quality or compound specific, rather than involving all taste sensation. Causes of taste dysfunction can be categorized as either drug/toxin effects or, less commonly, disease effects such as via periodontal disorders, neurologic disorders, nutritional disorders, infections, or endocrine disorders. Hedgehog inhibitor drugs, used to treat basal cell carcinoma, result in taste loss that can recover upon cessation of the drug. Radiation therapy for head and neck cancer can be another cause of taste dysfunction. Dysgeusia may (rarely) be a seizure manifestation. Iatrogenic injury to the chorda tympani during middle ear surgery can lead to gustatory complaints, such as a persistent metallic taste in the mouth, which gradually subside in most cases. Taste sensation has been shown to decline with age.

Burning mouth syndrome is a challenging problem. Patients complain of a burning oral sensation, often recurrent, with no apparent cause. Evaluation is aimed at eliminating any identifiable cause such as lesion, infection, or drug side effect. Management may include treating identifiable causes, removing offending medications, avoiding harsh toothpastes or mouthwashes, treating reflux, treating thrush, replacing ill-fitting dentures, general oral care and hydration, and trials of medications such as gabapentin for neuropathic pain. Also, modification of diet may be helpful.

◆ Evaluation

History and physical exam should be directed at determining whether the complaint truly relates to a taste disorder or (as is much more common) to an olfactory disorder. Objective testing for olfactory function such as the Smell Identification Test (Sensonics, Inc., Haddon Heights, NJ) can be helpful in this regard. Standardized objective testing for taste receptor function is not widely used. It is important to note that dysgeusia is usually described as one of the taste sensations (sweet, sour, etc.) or as metallic. If described as foul or rotten, this should alert the clinician to the possibility of an olfactory disorder.

◆ **Treatment Options**

If a taste disorder is identified, management is generally directed at removing or treating underlying causative factors and at supportive measures. Treating local infections such as oral thrush or periodontal disease may be helpful. Stopping possible offending systemic medications, if feasible, may be beneficial.

◆ **Outcome and Follow-Up**

Good oral care, hydration, nutrition, and xerostomia management with sialogogues are generally supportive measures. Radiation-induced taste dysfunction can recover gradually over many months but may be permanent.

